

Revision — percentages

The three questions, the multiplier and the reverse case, gathered before the control work.

LESSON 72–73

2 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 10 CONSOLIDATION

LESSON CLOCK

20'

25'

25'

8'

2

The methods on one page

20 min

Worked mixture

25 min

The errors that remain

25 min

A last check

8 min

Homework

2 min

LEARNING OBJECTIVES

- Choose the right method for each kind of percentage question.
- Work with increases, decreases and successive changes.
- Solve the reverse question reliably.
- Identify the errors that remain.

TEXTBOOK

National

pp. 206–210

Cambridge

Stage 7, pp. 96–104

Workbook

Revision 10

01

Explanation

The methods on one page

QUESTION	METHOD	EXAMPLE
a percentage of an amount	multiply by the decimal	$0.35 \cdot 240 = 84$
the whole from a part	divide by the decimal	$84 \div 0.35 = 240$
one amount as a percentage of another	$\frac{\text{part}}{\text{whole}} \cdot 100$	$\frac{84}{240} \cdot 100 = 35\%$
increase or decrease	$\cdot (1 \pm \frac{p}{100})$	$240 \cdot 1.35 = 324$
the original after a change	divide by the multiplier	$324 \div 1.35 = 240$
successive changes	multiply the multipliers	$1.2 \cdot 0.8 = 0.96$

SIX ROWS, AND THE WHOLE TOPIC

Every percentage question in the paper is one of these. Read the wording, pick the row, do the one operation it names.

Worked mixture

PROBLEM	ROW	ANSWER
18% of 350	first	63
63 is 18% of what?	second	350
63 as a percentage of 350	third	18%
increase 350 by 18%	fourth	413
after +18% it is 413: before	fifth	350
up 18%, then down 18%	sixth	-3.24%

ONE SET OF NUMBERS, ALL SIX QUESTIONS

Seeing 63, 350 and 18% arranged six ways is the fastest way to stop confusing the questions with each other.

The errors that remain

ERROR	LOOKS LIKE	CORRECT
dividing by the new value	$\frac{10}{50}$ for $40 \rightarrow 50$	$\frac{10}{40} = 25\%$
subtracting instead of dividing	$96\,000 - 19\,200$	$96\,000 \div 1.2$
adding successive percentages	$+20\% + 10\% = +30\%$	$+32\%$
a percentage over 100 rejected	“that must be wrong”	it is possible
profit taken on the selling price	$\frac{10\,000}{50\,000}$	$\frac{10\,000}{40\,000}$

A last check

ANSWER	SENSE CHECK
a part	smaller than the whole, unless the percentage is over 100
the whole from a part	larger than the part
after an increase	larger than before
after a decrease	smaller than before
a reverse question	multiply forwards and see

THE SENSE CHECK COSTS NOTHING

Most lost marks in this topic are answers that are obviously the wrong size. Ten seconds of checking recovers them.

02 Worked examples

EXAMPLE 1 Find 18% of 350, and then check that 63 is 18% of 350.

① $0.18 \cdot 350 = 63$

② $\frac{63}{350} = 0.18$

③ 18% ✓

The first and third questions, one after the other.

ANSWER

63 and 18%

EXAMPLE 2 After a rise of 18% a price is 413. Find the original.

① Multiplier 1.18.

② $413 \div 1.18$

③ $= 350$

Check: $350 \cdot 1.18 = 413$ ✓

ANSWER

350

EXAMPLE 3 A price goes up 18% and then down 18%. What is the net change?

① $1.18 \cdot 0.82$

② $= 0.9676$

③ A fall of 3.24%.

Never zero.

ANSWER

−3.24%

03 Interactive model

Write 63, 350 and 18% on the board and ask the class to invent all six questions; they revise the topic by writing it rather than reading it.

Which row of the table?

18% of 350:

63

350

413

18

63 is 18% of:

11.34

350

413

81

63 as a percentage of 350:

18%

56%

5.6%

180%

350 increased by 18%:

368

413

287

63

After +18% it is 413; before:

339

350

487

395

Up 18% then down 18%:

0%

-3.24%

+3.24%

-36%

From 40 to 50 the base is:

40

50

10

90

A profit percentage uses:

the selling price

the cost price

the profit

either

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Multiplier	Ko'paytuvchi	Множитель
Increase	Ortish	Увеличение
Decrease	Kamayish	Уменьшение
Original value	Boshlang'ich qiymat	Первоначальное значение
Percentage change	Foizli o'zgarish	Процентное изменение
Successive	Ketma-ket	Последовательный
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A percentage of an amount:

multiply

divide

add

subtract

The whole from a part:

multiply

divide

add

subtract

The original after an increase:

multiply by the multiplier

divide by the multiplier

subtract the percentage

add it back

Successive changes:

add

multiply

cancel

subtract

Percentage change divides by:

the new value

the old value

100

the change

An answer larger than the whole is:

always wrong

possible

impossible

a decimal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *18% of 350*

ANSWER *63*

2. *63 is 18% of*

ANSWER *350*

3. *63 as a percentage of 350*

ANSWER *18%*

4. *350 up 18%*

ANSWER *413*

5. *350 down 18%*

ANSWER *287*

6. *The multiplier for +18%*

ANSWER *1.18*

7. *The multiplier for -18%*

ANSWER *0.82*

Medium · the standard question

7 PROBLEMS

1. After +18% it is 413: before

ANSWER 350

2. After -18% it is 287: before

ANSWER 350

3. Up 18% then down 18%

ANSWER -3.24%

4. From 40 to 50

ANSWER +25%

5. From 50 to 40

ANSWER -20%

6. Bought 40 000, sold 50 000

ANSWER 25% profit

7. 12.5% of 480

ANSWER 60

Hard · stretch and reason

7 PROBLEMS

1. A price after 30% off is 147 000: the original

ANSWER 210 000

2. Up 25% then down 20%

ANSWER No change

3. Two rises of 15% from 200

ANSWER 264.5

4. 72 as a percentage of 60

ANSWER 120%

5. A salary of 3 000 000 after 12% rise and 5% tax

ANSWER 3 192 000

6. Which loses more: –10% twice, or –20% once?

ANSWER –20% once, at 0.8 against 0.81

7. Why check the size of a percentage answer?

ANSWER Most lost marks are answers of obviously the wrong size

07 Homework

Homework — 5 tasks

Name the row of the method table before every calculation.

- Find 22% of 450 and write 99 as a percentage of 450.
- Increase 450 by 22% and decrease it by 22%.
- After a rise of 22% a price is 549. Find the original.
- A price rises 30% and then falls 30%. Find the net change.
- A trader buys at 80 000 and sells at 92 000. Find the profit percentage.